



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

IMPORTANT: This course is for the Double Major pathway only. It should not be offered for the other four pathways.

Programme	B. Sc. Physics Honours				
Course Title	ELEMENTS OF MODERN PHYSICS				
Type of Course	Core in Major				
Semester	I or II (depending upon the batch in the Double Major)				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Higher secondary Physics				
Course Summary	The course integrates key principles of modern physics, including the Special Theory of Relativity, wave-particle duality, and the Bohr Atom Model, to provide students with a comprehensive understanding of fundamental concepts and their applications in diverse scientific fields.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the principles of the Special Theory of Relativity	Comprehension	Conceptual	Written exams, quizzes
CO2	Explain the dual nature of particles and waves	Comprehension	Conceptual	Problem sets, essays
CO3	Apply relativistic principles to solve problems	Application	Procedural	Problem-solving exams, simulations
CO4	Analyse experimental evidence supporting wave-particle duality	Analysis	Conceptual	Laboratory reports, case studies

CO5	Compare and contrast classical and quantum mechanical models	Analysis	Conceptual	Research papers, presentations
CO6	Critically evaluate the limitations of the Bohr atom model	Evaluation	Conceptual	Research projects, discussions
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	THE SPECIAL THEORY OF RELATIVITY		16	22
	1	Classical Relativity	1	
	2	The Michelson – Morley Experiment	1	
	3	Einstein’s Postulates and Its Consequences– Relativity of Time, Relativity of Length, Relativistic Velocity Addition	4	
	4	The Lorentz Transformation and Derivations of Relativistic Effects from Lorentz Transformations	2	
	5	Length Contraction, Velocity Transformation, Time Dilation, Simultaneity and Clock Synchronization	2	
	6	The Twin Paradox	1	
	7	Relativistic Dynamics – Relativistic Momentum	2	
	8	Relativistic Kinetic Energy, Total Energy and Rest Energy	2	
	9	Experimental Tests of Special Relativity	1	
	Sections 2.1 –2.7, 2.9 of chapter 2 of Book 1			
II	THE PARTICLE – LIKE PROPERTIES OF ELECTROMAGNETIC RADIATION		10	16
	10	Review of Electromagnetic Waves, Interference and Diffraction, Crystal Diffraction of X-Rays	2	
	11	The Photoelectric Effect	2	

	12	Thermal Radiation	2	
	13	The Compton Effect	2	
	14	Other Photon processes	1	
	15	Particles or Waves	1	
	Sections 3.1 – 3.6 of chapter 3 of Book 1.			
III	THE WAVE – LIKE PROPERTIES OF PARTICLES		10	16
	16	De Broglie’s Hypothesis	1	
	17	Experimental Evidences for De Broglie waves	3	
	18	Uncertainty Relationships for Classical waves	1	
	19	Heisenberg Uncertainty Relationships	2	
	20	Wave Packets and the Motion of a Wave Packet	2	
	21	Probability and Randomness, and the Probability Amplitude	1	
	Sections 4.1 – 4.7 of chapter 4 of Book 1			
IV	THE RUTHERFORD – BOHR MODEL OF THE ATOM		9	16
	22	Basic Properties of Atoms ,The Rutherford Nuclear Atom – Rutherford Scattering Formula and Its Experimental Verification – The Closest Approach of a Projectile to the Nucleus	2	
	23	Line Spectra	1	
	24	The Bohr Model	3	
	25	The Franck – Hertz Experiment	1	
	26	The Correspondence Principle	1	
	27	The Failure of the Bohr Model	1	
	Sections 5.1 – 5.8 of chapter 5 of Book1. Excluded: sections 5.2.1, 5.3.1, derivation of Rutherford scattering formula			
V	PRACTICALS		30	
	Conduct any 5 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 6 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			

1	Determination of Plank's constant using LEDs - Observe the turn-on voltage, V_0 of LEDs and calculate the value of h. Use at least 4 different colors of LED (with transparent casing) - Plot $\frac{1}{\lambda} - V_0$ graph using Python, fit a straight line to get the slope and estimate the value of h. - Calculate the %error. - Programmable voltage source of ExpEYES may be used to find the turn-on voltage.		
2	Continuous and line spectra- Determination of the wavelengths and photon energy. - Familiarize the initial adjustments and measurements in the spectrometer. - Mount the grating at normal incidence on the spectrometer. - Determine the wavelengths of the sodium vapor lamp and calculate the associated photon energy. - Determine the approximate range of the wavelengths of the continuous spectrum of incandescent/white LED lamp or any one coloured LED and calculate the associated photon energy. - The readings of the first order spectrum will be enough. Number of lines/m of the grating can be given.		
3	Mercury spectrum- Determination of wavelength and photon energy. - Determine wavelength of any four prominent lines and associated photon energy of the mercury spectrum using a spectrometer with grating at normal incidence. - The readings of the first order spectrum will be enough. Number of lines/m of the grating may be given.		
4	Hydrogen spectrum - Determination of wavelengths and calculation of the Rydberg's constant. - Determine the wavelengths and photon energy in eV of the prominent lines of the Balmer series of the Hydrogen spectrum using a spectrometer with grating at normal incidence. - Calculate the Rydberg's constant and estimate the % error. - The readings of the first order spectrum will be enough. Number of lines/m of the grating may be given.		
5	Wave Packets - Analysis of beats in sound. - The experiment is intended to understand the concept of wave packet, phase and group velocities. - Generate sounds waves of two near frequencies using smartphone/ExpEYES/Function generator and the		

		superimposed wave can be recorded and analysed using smartphone/ExpEYES/CRO ● Change the separation between the frequencies and compare the results with the theoretical values. ● https://expeyes.in/experiments/sound/beats.html ● Multi Tone generator and Audio scope tools of Phyphox may be used https://phyphox.org/experiment/tone-generator/		
6	Analysis of Hydrogen spectra using the Tracker Video Analysis tool. ● Calibrate the video of the Hydrogen spectra in the Tracker tool using two laser wavelengths/lines of mercury spectra. ● Plot the intensity profile, find the prominent wavelengths of the Balmer series and calculate the Rydberg's constant. ● Estimate the %error. ● Pre recorded video of the Hydrogen spectra can be used. ● https://physlets.org/tracker/. ● https://www.youtube.com/watch?v=UCCPkJpUQEw			
7	Black body spectrum of Sun -Estimation of surface temperature using the Tracker Video Analysis tool. ● Calibrate the video of the solar spectra in the Tracker tool using two laser wavelengths/lines of mercury spectra. ● Plot wavelength vs intensity, get λ_{max} and using Wein's law calculate the surface temperature. ● Pre recorded video of the solar spectra can be used.			
8	Verification of Wein's displacement law and Stefan's law using incandescent bulb. ● Calibrate the video of the spectra of the incandescent bulb in the Tracker tool using two laser wavelengths/lines of mercury spectra. ● Plot wavelength vs intensity and note λ_{max}. ● Repeat the experiment by increasing the operating voltage of the incandescent bulb(hence increasing the temperature of the source) ● From the plots, verify the Wein's displacement law and Stefan's law.			
9	Black body radiation- total energy output. ● Plot Planck's radiation formula. ● Evaluate the area under the curve and x- axis(total radiance over all wavelengths) by numerical integration and hence verify Stephan's law			

Books and References:

1. Modern Physics (Fourth Edition, an Indian Adaptation) by Kenneth S. Krane (Book 1)
2. <https://phyphox.org/>
3. <https://physlets.org/tracker/>

4. <https://expeyes.in/>
5. Modern Physics for Scientists and Engineers" by John Morrison
6. Concepts Of Modern Physics By Arthur Beiser
7. Modern Physics by Raymond A. Serway
8. Modern physics by Randy Harris

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	0	0	0	0	0	2	0	0	0	0	0	0
CO 2	0	3	2	0	0	0	0	2	0	0	0	0	0
CO 3	0	0	3	2	0	0	0	0	2	0	0	0	0
CO 4	0	0	0	3	2	0	0	0	0	2	0	0	0
CO 5	0	0	0	0	3	2	0	0	0	0	2	0	0
CO 6	0	0	0	0	0	3	0	0	0	0	0	2	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	